

EXPERIMENT NO. 4

Date: _____

TITLE

NumPy 2D Array – Branch Sales Analysis

AIM

To store sales data in a 2D array and compute branch/day-wise totals, averages and the highest-selling branch using NumPy.

OBJECTIVE

To use a 2D NumPy array for branch sales data and perform indexing, slicing and visualization to find weekly totals, branch averages, day-wise totals and the top branch.

THEORY

Sales matrix (3 branches $\times$ 7 days):

$$S = \begin{bmatrix} s_{11} & \cdots & s_{17} \\ s_{21} & \cdots & s_{27} \\ s_{31} & \cdots & s_{37} \end{bmatrix}$$

Indexing and slicing:

$$S[0], S[1, 3:], S[:, 0:3]$$

Weekly total per branch and grand total:

$$W_i = \sum_{j=1}^7 s_{ij} \quad (\text{axis} = 1) \quad T = \sum_{i,j} s_{ij}$$

Branch average and day-wise total:

$$\bar{B}_i = \frac{1}{7} \sum_{j=1}^7 s_{ij} \quad D_j = \sum_{i=1}^3 s_{ij} \quad (\text{axis} = 0)$$

Top branch:

$$k = \arg \max_i W_i$$

PROCEDURE

1. Store the sales data of 3 branches over 7 days in a 2D NumPy array and display it.
2. Display the 1st branch sales, Mon–Wed sales and 2nd-branch sales from Thursday.
3. Find weekly total per branch, grand total, branch-wise average and day-wise total.
4. Find the branch with the highest sales, display all results and plot the daily-sales bar, pie, line and scatter charts.

DATASET DESCRIPTION

Dataset Name: Branch sales (entered in program)

Source: 3 branches $\times$ 7 days (Mon–Sun), synthetic figures

Number of Samples: 21 entries Number of Features: 7 days

Target / Class: Not applicable (aggregation operations)

Description: B1=[1200, 1350, 1100, 1450, 1600, 2000, 1800]; B2=[900, 1100, 950, 1200, 1300, 1700, 1500]; B3=[1500, 1600, 1400, 1750, 1900, 2200, 2100].

PROGRAM

```

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]

sales = np.array([
    [1200, 1350, 1100, 1450, 1600, 2000, 1800],
    [900, 1100, 950, 1200, 1300, 1700, 1500],
    [1500, 1600, 1400, 1750, 1900, 2200, 2100],
])

print("Complete sales data (branches x days):")
print(sales)

print("1st branch sales =", sales[0])

print("Mon-Wed sales:")
print(sales[:, 0:3])

print("2nd branch from Thursday =", sales[1, 3:])

print("Weekly total per branch =", sales.sum(axis=1))
print("Grand total =", sales.sum())

print("Branch-wise average =", np.round(sales.mean(axis=1), 2))

print("Day-wise total =", sales.sum(axis=0))

top = int(sales.sum(axis=1).argmax())
print("Highest sales: branch", top + 1)
x = np.arange(len(days))
w = 0.25
plt.figure(figsize=(7, 4.5))
plt.bar(x - w, sales[0], width=w, label="Branch 1")
plt.bar(x, sales[1], width=w, label="Branch 2")
plt.bar(x + w, sales[2], width=w, label="Branch 3")
plt.xticks(x, days)
plt.xlabel("Day"); plt.ylabel("Sales"); plt.title("Daily Sales per Branch")
plt.legend(); plt.tight_layout(); plt.savefig("exp4_sales.png", dpi=150)
plt.figure(figsize=(5, 4))
L = ["Branch 1", "Branch 2", "Branch 3"]
plt.pie(sales.sum(axis=1), labels=L, autopct="%1.1f%%")
plt.title("Branch Share of Total Sales"); plt.tight_layout()
plt.savefig("exp4_pie.png", dpi=150)
plt.figure(figsize=(7, 4.5))
plt.plot(days, sales.sum(axis=0), marker="o", linewidth=2)
plt.xlabel("Day"); plt.ylabel("Total sales")
plt.title("Day-wise Total Sales Trend")
plt.tight_layout(); plt.savefig("exp4_line.png", dpi=150)
plt.figure(figsize=(7, 4.5))
plt.scatter(range(7), sales[0], s=40, label="Branch 1")
plt.scatter(range(7), sales[1], s=40, label="Branch 2")
plt.scatter(range(7), sales[2], s=40, label="Branch 3")
plt.xlabel("Day index"); plt.ylabel("Sales"); plt.title("Daily Sales Scatter per Branch")
plt.legend(); plt.tight_layout(); plt.savefig("exp4_scatter.png", dpi=150)
print("saved exp4_sales.png exp4_pie.png exp4_line.png exp4_scatter.png")

```

OUTPUT

Complete sales data (branches x days):

```
[[1200 1350 1100 1450 1600 2000 1800]
 [ 900 1100  950 1200 1300 1700 1500]
 [1500 1600 1400 1750 1900 2200 2100]]
```

1st branch sales = [1200 1350 1100 1450 1600 2000 1800]

Mon-Wed sales:

```
[[1200 1350 1100]
 [ 900 1100  950]
 [1500 1600 1400]]
```

2nd branch from Thursday = [1200 1300 1700 1500]

Weekly total per branch = [10500 8650 12450]

Grand total = 31600

Branch-wise average = [1500. 1235.71 1778.57]

Day-wise total = [3600 4050 3450 4400 4800 5900 5400]

Highest sales: branch 3

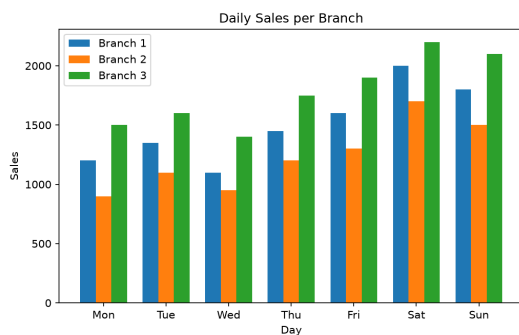


Fig. 1: Daily sales per branch.

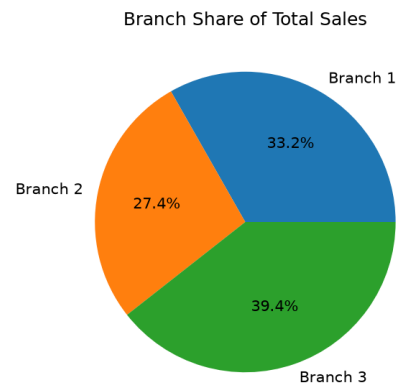


Fig. 2: Branch share of total sales.

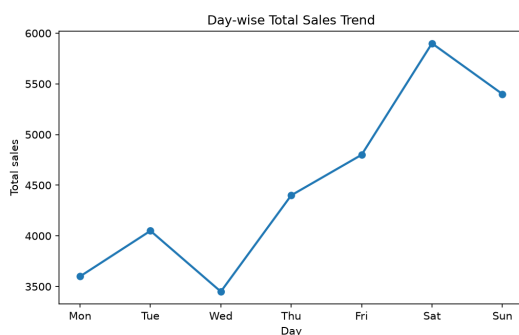


Fig. 3: Day-wise total sales trend.

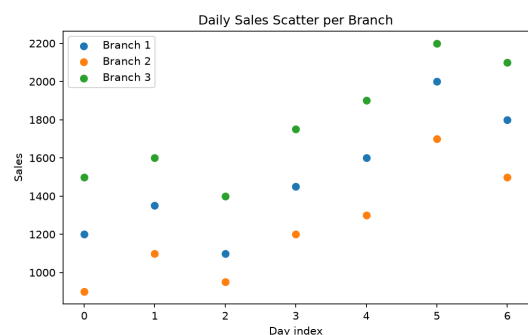


Fig. 4: Daily sales scatter per branch.

RESULT

The above program was implemented and executed successfully using Python, and the corresponding output was obtained and verified. The results of the **NumPy data visualization** were analyzed and found to be satisfactory, thereby achieving the desired objective of the experiment.

Faculty Signature: _____

Date: _____